

Math 103

Class 7

1. Variance and Standard Deviation (of a sample)

Tells how "spread-out" the data is. (0 or above)

A variance of zero says "all data entries are the same".

$$\text{Variance} = S^2 = \frac{\sum(x - \bar{x})^2}{n-1} \quad (\text{good formula if } x \text{ is a whole \#.})$$

Example:

<u>x</u>	<u>x - $\bar{x}$</u>	<u>(x - $\bar{x}$)²</u>	
7	-1	1	$\bar{x} = 8$
6	-2	4	
6	-2	4	
7	-1	1	$s^2 = \frac{40}{9 - 1} = 5$
12	4	16	
9	1	1	
6	-2	4	$s = \sqrt{5}$
11	3	9	
8	0	0	$= 2.236$
		<u>40</u>	$= 2.2$

Shortcut formula

$$\text{Variance} = S^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}, \text{ we will use this formula}$$

Example:

<u>x</u>	<u>x²</u>	
7	49	$s^2 = 616 - \frac{(72)^2}{9}$
6	36	
6	36	
7	49	
12	144	$= 616 - \frac{5184}{9}$
9	81	
6	36	
11	121	
<u>8</u>	<u>64</u>	$= \frac{616 - 576}{8}$
$\Sigma x = 72$	$\Sigma x^2 = 616$	
		$= \frac{40}{8}$
		$= 5$
		$s = 2.2$

2. Practice:
p. 122:2

45	23	52	33	29	range = 29
					$\Sigma x = 182$
					$\Sigma x^2 = 7188$
					$s^2 = 140.90$
					$s = 11.87$
					$\bar{x} = 36.4$

Same example: Find the "coefficient of variation"

$$\text{C.V.} = s/\bar{x} \bullet 100 \quad s = \text{st. dev.}$$

$$\bar{x} = \text{mean}$$

$$\begin{aligned} \text{C.V.} &= 11.87/36.4 \bullet 100 \\ &= 32.6\% \end{aligned}$$

The "coefficient of variation" expresses the stand. dev. as a percentage of the sample (or pop.) mean.

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Same example: apply Chebyshev's Theorem (p. 120)

$$1 - 1/k^2 \quad k = \text{the number of standard dev.}$$

At least 75% of the data will fall in the interval
from $\mu - 2\sigma$ to $\mu + 2\sigma$

$$\begin{array}{rcl} 36.4 - 2(11.87) & \text{to} & 36.4 + 2(11.87) \\ 36.4 - 23.74 & \text{to} & 36.4 + 23.74 \\ 12.66 & \text{to} & 60.14 \end{array}$$

3. Practice: p.123:5a

56 85 52 13 39

range = 72
Mean = 49
 $s^2 = 687.49$
 $s = 26.22$
C.V. = 53.5%
Chebyshev -3.4 to 101.44

4. Practice: p. 123:5b

24 53 60 69 18

range = 51
Mean = 44.8
 $s^2 = 508.50$
 $s = 22.55$
C.V. = 50.3%
Chebyshev -.3 to 89.9